

Operating System Security Assessment and Security Plan

Completed as part of the IBM SkillsBuild Cybersecurity
Certificate Program

Overview

This project involved evaluating operating system security controls and developing a security plan to improve system protection. The assessment focused on software updates, device and port security, event logging, authentication controls, and backup strategies.

Objectives

- Evaluate operating system update settings and security posture.
- Assess active ports and connected devices for potential security risks.
- Determine the availability and usefulness of event logging.
- Review authentication controls protecting sensitive information.
- Evaluate backup practices and recovery readiness.

Skills Demonstrated

- Operating System Security
- Security Hardening
- Access Control

- Event Logging and Monitoring
- Risk Assessment
- Backup and Recovery Planning
- Cybersecurity Best Practices

Activities Performed

- Reviewed operating system update configuration and patch management practices.
- Evaluated device and port exposure to identify potential attack surfaces.
- Assessed logging and monitoring capabilities available within the operating system.
- Examined authentication methods used to protect sensitive information.
- Reviewed backup procedures and recovery preparedness measures.

Outcome

Developed a security plan containing recommendations to strengthen operating system security, improve monitoring capabilities, enhance authentication controls, and improve data resilience through backup strategies.

Lessons Learned

- Regular operating system updates are critical for reducing security vulnerabilities.
- Unnecessary ports and devices can increase the attack surface of a system.

- Event logging plays an important role in security monitoring and incident investigation.
- Strong authentication mechanisms improve protection of sensitive information.
- Reliable backup procedures help organizations recover from data loss and cyber incidents.

Prepared by:

Daniel Tetenu

Based on work completed as part of the IBM SkillsBuild Cybersecurity Certificate Program.

